

BUSINESS STORYTELLING

Project Title:

Retail & E-Commerce Sales Performance Analysis

1. Objective

The objective of this project was to analyze the retail and e-commerce sales data to understand overall business performance, identify key trends, evaluate customer purchasing behavior, measure product profitability, and provide data-driven recommendations through an interactive Power BI dashboard.

2. Business Background

Retail businesses generate a large volume of transactional data every day. Without proper analysis, it becomes difficult to identify which products perform well, which regions generate the highest revenue, how customer purchasing behavior changes over time, and how business profitability is affected by discounts and product returns.

To solve these challenges, this project analyzes historical sales data and transforms it into meaningful business insights using Microsoft Excel, Power Query, DAX, and Power BI.

3. Business Problem

The business wanted answers to several important questions:

- How are sales changing over time?
- Which product categories and products generate the highest sales and profit?
- Which customer segments contribute the most revenue?
- Which regions perform the best?
- Which payment methods are preferred?
- How do discounts affect profitability?
- Why are products being returned?
- What is the overall business performance?

Answering these questions helps management make informed business decisions.

4. Project Approach

The project was completed in the following phases:

Phase 1

Data Collection

Collected Orders and Returns datasets containing sales transactions and return information.



Phase 2

Data Understanding

Studied the dataset structure, identified data quality issues, and understood business requirements.



Phase 3

Data Quality Assessment

Verified missing values, duplicate records, invalid values, incorrect data types, and business rule violations.



Phase 4

Power Query Data Cleaning

Performed data cleaning and transformation by:

- Removing duplicate records.
- Handling missing values.
- Correcting invalid Quantity and Discount values.
- Standardizing Category values.
- Applying Trim and Clean.
- Assigning correct data types.



Phase 5

Exploratory Data Analysis (EDA)

Performed detailed analysis to answer twelve business questions covering:

- Monthly sales trends
- Category performance
- Product performance
- Brand performance
- Customer segments
- Region and state analysis
- Shipping modes
- Payment methods
- Discount analysis
- Product returns
- Overall KPIs



Phase 6

Data Modeling

Created relationships between Orders and Returns tables to build an efficient data model.



Phase 7

DAX Measures

Developed dynamic business calculations including:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity Sold
- Returned Orders
- Return Rate



Phase 8

Dashboard Development

Designed an interactive Power BI dashboard containing:

- KPI Cards
- Interactive Slicers
- Line Chart
- Column Charts
- Bar Charts
- Donut Charts
- Drill-Down Analysis



Phase 9

Business Insights

Identified important business trends including:

- Best-performing categories
- Best-selling products
- Most profitable regions

- Customer purchasing behavior
- Discount impact
- Return patterns



Phase 10

Business Recommendations

Prepared practical recommendations to improve sales, profitability, operational efficiency, and customer satisfaction.

5. Business Value

The dashboard enables management to:

- Monitor overall business performance.
- Identify high-performing products.
- Compare category performance.
- Analyze regional sales.
- Understand customer purchasing behavior.
- Monitor returns.
- Track important KPIs.
- Make faster data-driven decisions.

6. Project Outcome

The project successfully transformed raw retail transaction data into meaningful business information.

The final Power BI dashboard provides an interactive platform where users can monitor sales performance, analyze profitability, identify trends, and evaluate business performance through dynamic filters and visualizations.

7. Conclusion

This project demonstrates the complete end-to-end Data Analytics process, including data collection, data quality assessment, data cleaning, exploratory data analysis, data modeling, DAX calculations, dashboard development, business insight generation, and business recommendations.

The dashboard supports informed decision-making by presenting complex business data in a simple, interactive, and visually meaningful format, enabling organizations to improve operational efficiency and achieve better business outcomes.